

Infection Control & Hospital Epidemiology - Review Article

Article Information

Antibiotic Stewardship in Acute Care Hospitals:

Current Challenges and Future Directions

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Abstract

Antibiotic resistance accounts for 2.8 million infections and 35,000 deaths annually in the US. 30-50% of hospital antibiotics are unnecessary or inappropriate. This review examines current challenges in ASP implementation and proposes frameworks for next-generation stewardship programs.

Case Example - Fictional Hospital

Memorial Community Hospital - ASP Implementation

Baseline Data (Pre-ASP):

- Total antibiotic use: 852 DOT/1000 patient-days
- Broad-spectrum ratio: 42%
- C. difficile rate: 8.2/10,000 patient-days
- MRSA bacteremia: 1.8/10,000 patient-days

Interventions Implemented:

1. Prospective audit with pharmacist intervention
2. Electronic clinical decision support
3. IV-to-oral conversion protocol
4. Cultures-before-antibiotics requirement

Results After 12 Months:

- Total antibiotic use: 687 DOT/1000 patient-days (-19%)
- Broad-spectrum ratio: 28% (-14%)
- C. difficile rate: 4.1/10,000 (-50%)
- MRSA bacteremia: 1.1/10,000 (-39%)
- Cost savings: \$340,000 annually

Key Strategies

1. Prospective audit and feedback
2. Formulary restriction and preauthorization
3. Clinical decision support systems
4. De-escalation protocols
5. IV-to-oral conversion
6. Duration optimization
7. Diagnostic stewardship